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RESEARCH ARTICLE

Pseudo-Zero-Gap Flow-Type Aqueous Zn-CO_2 Batteries

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ABSTRACT

Aqueous Zn-CO_2 batteries (AZCBs) generate electricity and convert CO_2 into valuable products simultaneously, while their development is limited due to poor mass transport, low reactivity, etc. Therefore, we develop AZCBs featuring flowing CO_2 gases and electrolytes, gas-diffusion cathodes, and especially a pseudo-zero interelectrode distance. By employing carbon-coated bismuth oxides (Bi_2O_3/C) as the cathode in a pseudo-zero-gap AZCB, a record-high peak power density of 67.0 mW cm^{-2} @current density of 117.5 mA cm^{-2} is achieved, together with excellent cycling performance over 600 cycles. Specifically, the near-zero distance greatly promotes hydroxide ion transport and CO_2 diffusion by establishing an enhanced concentration gradient at the reaction interface of CO_2 electroreduction. Theoretical simulations further indicate preferred formate production is due to the lower energy barrier compared to the CO pathway, because of the favorable *O@Bi adsorption via the *OCHO intermediate. Significantly, the developed AZCBs operate efficiently and stably in a simulated Martian atmosphere, which is promising for future Human exploration of Mars, providing a potential solution for simultaneous energy generation and feedstock production in the challenging environment.

1 | Introduction

Fast depletion of fossil fuels by modern industry has resulted in severe global greenhouse effect, limited energy conservation, etc [1, 2]. To mitigate such environmental and energy crises [3, 4], the CO_2 battery could be a promising technology since it converts CO_2 into valuable chemicals/fuels and generates electricity simultaneously [5]. Furthermore, due to high CO_2 concentration of the Martian atmosphere (95.3% CO_2, 2.7% N_2, 1.6% Ar, etc.), it might also be a solution to generate carbon-based feedstocks and electric energy for Mars exploration using its own resources [6, 7]. Among various CO_2 batteries, aqueous Zn-CO_2 batteries (AZCBs) stand out because of their low cost, environmental friendliness, renewability, safety, etc [8, 9]. Especially, Zn metals can endure reactive aqueous electrolytes, generating value-added carbon products by electrochemical CO_2 reduction reaction (eCO_2RR) during battery discharging [10]. However, widespread development of AZCBs is still limited by several stumbling bottle-neck problems such as low discharge potential and current density, small power density, poor energy efficiency, etc., constraining their widespread applications [11].

In a typical AZCB, the anode proceeds Zn/Zn^{2+} redox reactions (Discharging: $Zn \rightarrow Zn^{2+} + 2e^-$, $Zn^{2+} + 4OH^- \rightarrow Zn(OH)_4 2^-$; Charging: $Zn(OH)_4 2^- + 2e^- \rightarrow Zn + 4OH^-$), during which the hydroxide ion transport plays a critical role. As for the cathode, eCO_2RR (Discharging: for example, $CO_2 + H_2O + 2e^- \rightarrow HCOO^- + OH^-$) and e$CO_2$RR product (Charging: $HCOO^- + OH^- \rightarrow CO_2 + 2e^- + 2H_2O$) or water oxidation reaction (Charging: if

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gaseous eCO_2 RR products generated, $4OH^- \rightarrow O_2 + 4H_2O + 4e^-$) proceed, that corresponding electrocatalysts, CO_2 diffusion, ion transport matter [12]. Currently, most research focuses on eCO_2RR electrocatalysts to regulate discharging products and battery performances, while battery power density is still limited to less than 10 mW cm^{-2} due to low discharge potentials and small current densities (as shown in Table S1) [13, 14]. This demonstrates that mass transport, such as CO_2, OH^-, etc., also needs to be addressed to break the bottleneck of battery performance, which might be achieved by battery configuration optimization.

Hitherto, the most widely explored configuration for CO_2 battery is the conventional H-type batch device, rendering its performance far from practical applications [15]. In which, the long distance between cathodes and anodes results in inefficient ion and CO_2 transport, so that CO_2 first dissolves in aqueous electrolytes and then slowly diffuses to the cathode, taking part in eCO_2RR. It is known that cathodic eCO_2RR performance is highly dependent on CO_2 transport behavior, which could be greatly promoted by gas-diffusion electrodes (GDEs) [16]. Therefore, flow-type CO_2 batteries are developed, equipped with circulating electrolytes, flowing CO_2 gas, and GDEs, aiming for promoting mass transport and thus cathodic eCO_2RR kinetics [17]. As a result, the battery performance, such as peak power density, could be improved from less than 10 to 15.9 mW cm^{-2} [18], even 20.2 mW cm^{-2} [19]. However, such performance is still far from applications, as compared with mature Zn-air batteries with peak power densities of over 100 mW cm^{-2} [20]. The possible reasons might be such loose configurations of AZCBs, no matter H-type or Flow-type one, resulting in large interelectrode distances and thus greatly compromised OH^- transport and CO_2 diffusion efficiency during battery working, leading to low current density, small power density, additional energy loss, etc [21].

In line with the above hypothesis, here we innovatively propose a pseudo-zero-gap AZCB system, which not only utilizes GDEs to promote CO_2 diffusion but also reduces mass-transport resistance by minimizing the distance between anodes and cathodes. The as-fabricated AZCBs display superior discharging performance with a record-high peak power density of 67.0 mW cm^{-2} and current density of 117.5 mA cm^{-2}, several times of reported data. Especially, the ion and CO_2 concentration together with current density distribution and power density of AZCBs with various battery configurations are simulated by COMSOL using a two-dimenisonal (2D model. Furthermore, the cathodic eCO_2RR to formate reactivity is regulated by optimizing carbon-coated Bi_2O_3 (Bi_2O_3/C) electrocatalysts through annealing a Bi-hexahydroxytriphenyl (Bi-HHTP) MOF at different temperatures. Additionally, the eCO_2RR mechanism is investigated by density functional theory (DFT) calculations, confirming the Bi_2O_3/C catalyst prefers formate due to the largeenergy-barrier gap between the formate and carbon monoxide pathway. Significantly, the great performance of AZCBs in a simulated Martian atmosphere suggests it might be a highly promising energy and feedstock solution for future Mars exploration.

2 | Results and Discussion

2.1 | Battery Working Principles

During the discharging process of pseudo-zero-gap AZCB (Figure 1), CO_2 is purged into a gas chamber at the left side of cathode, and then transports through the cathode diffusion layer (CDL) to the cathode catalyst layer (CCL), where CO_2 and H_2O are reduced by electrons to formate ($HCOO^-$) and OH^- ions in an alkaline electrolyte (6 M KOH + 0.02 M $Zn(CH_3COO)_2$, Equation 1). Side reactions like the hydrogen evolution reaction (HER) might also happen, as Equation 2. The generated OH^- ions migrate through the porous glass fiber separator (Figure S1) to the Zn-plate anode, where Zn is oxidized to Zn^{2+} reacting with OH^-, forming $Zn(OH)_4{}^{2-}$ complexes, and releasing electrons to the external circuit (Equation 3). Simultaneously, $Zn(OH)_4{}^{2-}$ complexes would proceed dehydration to ZnO and H_2O in alkaline as Equation 4. Therefore, the total discharging reaction is as Equation 5.

During battery charging, $Zn(OH)_4{}^{2-}$ at the anode is reduced back to metallic Zn by electrons releasing OH^- ions (Equation 6), which travel through the porous glass fiber to the cathode, where they react with $HCOO^-$ via an oxidation process, producing CO_2 (Equation 7), contributing to a high open-circuit potential (OCP). Side reactions such as oxygen evolution reaction (OER) might also happen at the cathode during charging (Equation 8). So, the overall charging reaction is shown as Equation 9. Specifically, the pore size of the glass fiber separator (typically 0.7–2.7 μm) is significantly larger than the hydration radius of OH^- ions (0.3–0.4 nm), permitting the diffusion of OH^- ions (Figure S1). In addition, the electrolyte providing a large amount of OH^- ions could be supplemented by the anolyte travelling from the right side of a perforated Zn plate, as shown in Figure S2. Therefore, the ion exchange of this pseudo-zero-gap AZCB is completed by a delicate design of a porous glass fiber sitting between the cathode and anode, and a hollow Zn anode permitting in-time supply of OH^- from the electrolyte chamber.

During the discharge process, the following reactions happen:

Cathode:

$$CO_2 + H_2O + 2e^- \rightarrow HCOO^- + OH^-;\ \ E^\theta = -0.61\ \text{V vs. SHE} \tag{1}$$

Cathode side reaction:

$$2H_2O + 2e^- \rightarrow H_2 + 2OH^-;\ \ E^\theta = -0.82\ \text{V vs. SHE} \tag{2}$$

Anode:

$$Zn + 4OH^- \rightarrow Zn(OH)_4^{2-} + 2e^-;\ \ E^\theta_{Zn/Zn(OH)_4^{2-}} = -1.25\ \text{V. vs. SHE} \tag{3}$$

Anode side reaction:

$$Zn(OH)_4^{2-} \rightarrow ZnO + 2OH^- + H_2O \tag{4}$$

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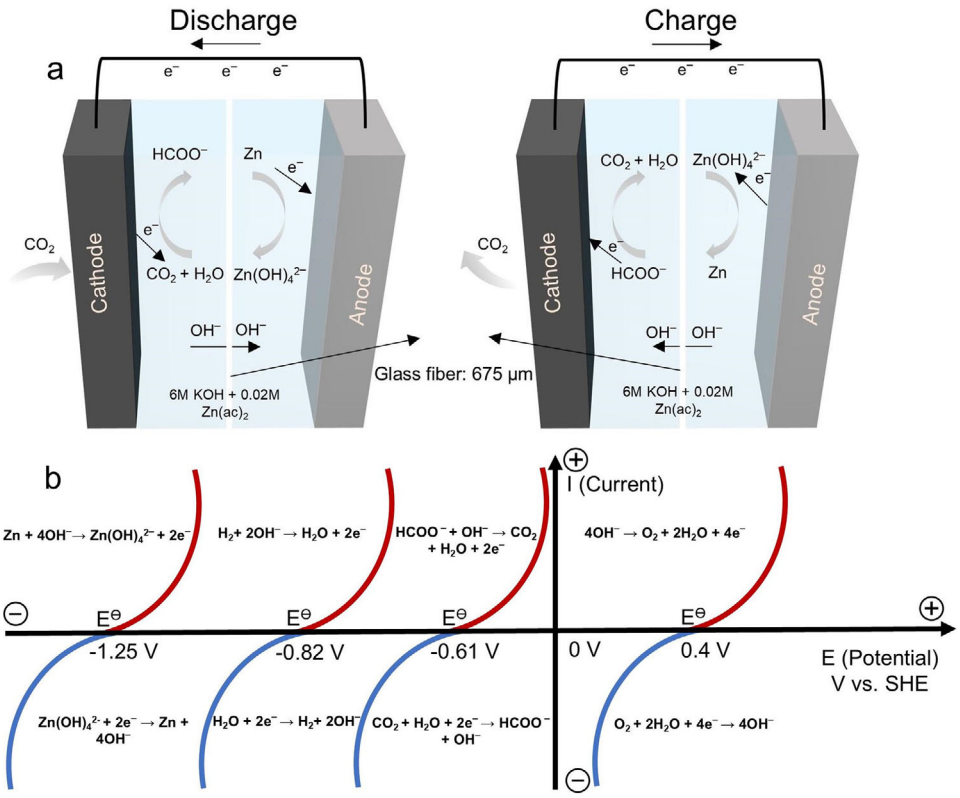

FIGURE 1 | (a) Battery reactions during discharge and charge at the cathode and anode; (b) The standard electrode potentials for battery reactions.

Therefore, the overall discharging reaction is as follows:

$$CO_2 + Zn + OH^- \rightarrow HCOO^- + ZnO; \cdot E_c = -0.61V \text{ vs. SHE} \quad (5)$$

During the charging process, reversible reactions happen:

Anode:

$$Zn(OH)_4^{2-} + 2e^- \rightarrow Zn + 4OH^-;\ E^{\theta}_{Zn/Zn(OH)_4^{2-}} = -1.25 \text{ V vs. SHE} \quad (6)$$

Cathode:

$$HCOO^- + OH^- \rightarrow CO_2 + H_2O + 2e^-;\ E^{\theta} = -0.61 \text{ V vs. SHE} \quad (7)$$

Cathode side reaction:

$$4OH^- \rightarrow O_2 + 2H_2O + 4e^-; \cdot\cdot E^{\theta} = 0.4V\text{vs. SHE} \quad (8)$$

Therefore, the overall charging reaction is as follows:

$$HCOO^- + Zn(OH)_4^{2-} \rightarrow CO_2 + Zn + 3OH^- + H_2O;\ E_a = -1.25 \text{ V vs. SHE} \quad (9)$$

In addition, based on the Nernst equation (please see 1.3.4 in the Supporting Information), and after correction under non-standard operating conditions, the designed pseudo-zero-gap AZCB exhibits a theoretical energy density of 235.1 Wh kg^{-1}, indicating great thermodynamic feasibility.

2.2 | Battery Configuration Optimization

As illustrated in Figure 2a–c, Figure S3, three types of AZCB configurations are fabricated and compared. A traditional H-type battery consists of a Zn anode and eCO_2RR cathode (with an interelectrode distance ~8.0 cm) separated by a bipolar membrane (BPM), where CO_2 is bubbled into the electrolyte, diffuses to the cathode surface through an aqueous electrolyte, and then reacts with protons and electrons generating eCO_2RR products (Figure 2a; Figure S3a). However, the low solubility of CO_2 in aqueous solutions (33 mM, 20 °C, 1 atm) together with its small diffusion coefficient (1×10^{-5} cm^2 s^{-1}), result in inefficient CO_2 transport. Furthermore, the interelectrode spacing is as large as 8.0 cm, leading to high ohmic and ion transport resistance. To promote mass transport, a flow-type battery is assembled (Figure 2b; Figure S3b) that a GDE-cathode facing catholyte and Zn anode facing anolyte are separated by a BPM, with an electrode gap of 2.013 cm (the thickness of catholyte, anolyte chamber, and BPM is 1.0, 1.0, and 0.013 cm, respectively). Besides, it incorporates an additional CO_2 chamber near the cathode GDE compartment, allowing CO_2 to diffuse to the CCL through a gas diffusion layer (GDL), promoting eCO_2RR. The catholyte and anolyte are continuously pumped through the battery, which benefits product separation and electrolyte replenishment. Therefore, flow-type AZCBs enhance CO_2 diffusion and ion transport by applying GDL and decreasing the interelectrode distance from 8.0 to 2.013 cm, as compared with the H-type battery. To further optimize the mass transport, we reduce the interelectrode gap by removing the catholyte chamber and separating the GDE-cathode and Zn anode only by a glass fiber (Whatman GF/D,

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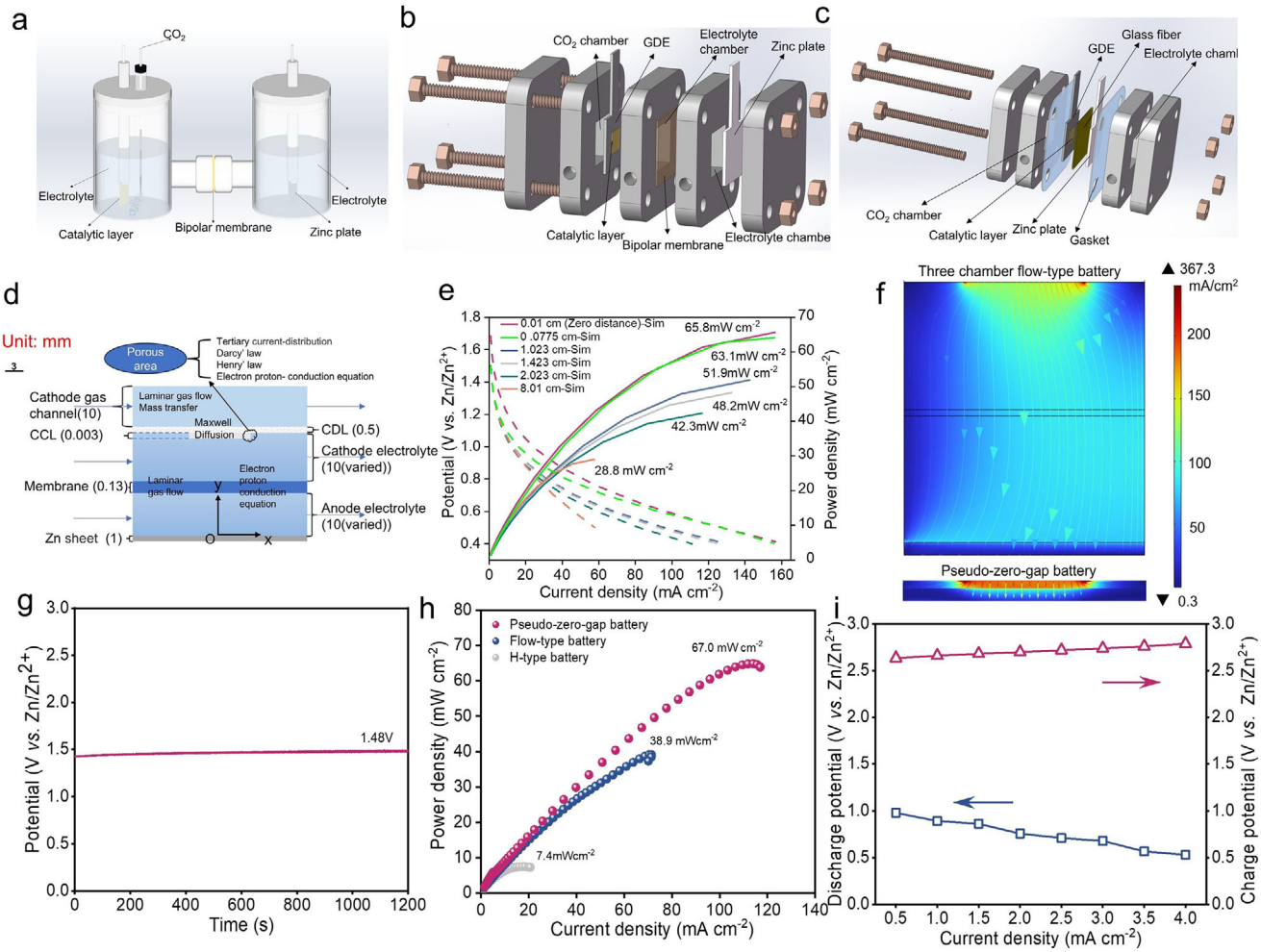

FIGURE 2 | (a) The H-type AZCB; (b) The three-chamber flow-type AZCB; (c) The pseudo-zero-gap AZCB; (d) 2D modelling configuration of AZCBs; (e) Experimental and simulated *I–V* and *I–P* curves of AZCBs with different interelectrode gaps; (f) Current flow and current density distribution in the electrolyte for the pseudo-zero-gap batteries (Bottom) and Flow-type batteries (Top); (g) OCP of the pseudo-zero-gap AZCB. The anolyte is 6 M KOH + 0.02 M $Zn(ac)_2$, and the catholyte is 6 M KOH. (h) The *J–P* curves of different AZCBs using the same cathode material; (i) Continuous charge and discharge curves of the pseudo-zero-gap AZCB.

with a thickness of 675.0 μm and an aperture of 2.7 μm, Figure 2c; Figure S3c). In such a device, the CO_2 chamber faces the cathode GDL, the anolyte faces the backside of the Zn plate, without the catholyte chamber to reduce the electrode gap, alleviate electrolyte overflow, and carbonate salt precipitation, etc. This design is considered to shorten the ion transport distance and thus minimize the total system resistance while retaining excellent CO_2 mass transport capability.

To testify above hypothesis, the battery performance is studied both theoretically and experimentally. First, COMSOL simulations are conducted using a 2D model, with the battery model being simplified with a Zn anode, anolyte chamber, catholyte chamber, separator, and CCL, CDL, gas flow channel at the cathode (Figure 2d; Figures S4 and S5). In which, the distance between the Zn anode and cathode (mainly CCL) is set as the primary variable, ranging from 8.01 (H-type), 2.023 (Flow-type), 1.083, 1.063, 1.043, 1.023, 0.0775 (pseudo-zero-gap), and 0.01 cm (zero distance). An additional 0.01 cm is assigned to the catholyte chamber during simulation to accelerate the convergence process and prevent computational failure while progressively reducing its thickness. Accordingly, the polarization curve (J-V) and power density plot with current density (J-P) are simulated, that the peak power density decreases from 65.8 mW cm^{-2} (0.01 cm) to 63.1 mW cm^{-2} (0.0775 cm), 51.9 mW cm^{-2} (1.023 cm), 48.2 mW cm^{-2} (1.423 cm) and 42.3 mW cm^{-2} (2.023 cm), and 28.8 mW cm^{-2} (8.01 cm), respectively (Figure 2e). It can be seen that the peak power density of AZCB with an actual interelectrode distance of 0.0675 cm is very close to that of zero distance, so the former is defined as a pseudo-zero-gap AZCB. Besides, the current density distribution of AZCBs with different interelectrode gaps is also simulated under a potential of 0.4 V vs Zn/Zn^{2+} (Figure 2f; Figure S6). It is evident that the battery with the smallest interelectrode spacing exhibits the highest current density, which decreases progressively as the electrode distance increases, that of the pseudo-zero-gap one close to 200 mA cm^{-2}.

Experimentally, using an efficient eCO_2RR catalyst (Bi_2O_3/C, which could produce HCOOH efficiently) to construct cathodes for AZCBs with as-proposed H-type, flow-type, and pseudo-zero-gap configurations, respectively, the actual battery performance is

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measured. As shown in Figure 2g, the assembled pseudo-zero-gap battery exhibits a stable OCP around 1.48 V in pure CO_2 atmosphere. Polarization curves of the battery charge and discharge process have also been tested, and then the power density is calculated (Figure 2h; Figure S7). Specifically, the peak power density of pseudo-zero-gap AZCB could reach 67.0 mW cm^{-2} at a current density of 117.5 mA cm^{-2} and a discharge potential of 0.57 V vs. Zn/Zn^{2+}, close to that of simulated data (63.1 mW cm^{-2}) and much higher than that of flow-type (37.2 mW cm^{-2} at 65.11 mA cm^{-2}), and H-type battery (7.4 mW cm^{-2} at 19.35 mA cm^{-2}). Thus far, the obtained value is the highest among reported data (Table S1) [22]. Furthermore, the discharge potential of this pseudo-zero-gap AZCB remains high level reaching 0.98 V at 0.5 mA cm^{-2}, still above 0.4 V at 4 mA cm^{-2} and only a low potential of 2.7 V is needed during galvanostatic charging at 4 mA cm^{-2} (Figure 2i; Figure S8), suggesting its excellent continuous charge and discharge performance.

2.3 | Origin of Advantageous AZCB Performance

Based on above results of AZCBs with different configurations, the pseudo-zero-gap one displays the best battery performance with the highest current density, largest peak power density, etc. While the underlying reason should be revealed, which is related to the system mass transport and local reactivity. Since gaseous CO_2 ($CO_{2,G}$) could not be adsorbed on catalytic metal surfaces under ambient conditions (1 atm, 293 K), dissolved CO_2 ($CO_{2,D}$) is considered as the real reactant during eCO_2RR, which is converted from $CO_{2,G}$ [23, 24]. Specifically, in AZCBs using GDEs (flow-type and pseudo-zero-gap battery), $CO_{2,G}$ first travels through the porous CDL (with a thickness of 45 μm) to CCL at a high diffusion coefficient (1×10^{-1} cm^2 s^{-1}), then dissolves in the electrolyte forming $CO_{2,D}$ which travels to the catalytic surface at a low diffusion coefficient (1×10^{-5} cm^2 s^{-1}, the travel distance varies from nanometers to micrometers depending on the device configuration), and lastly been consumed by eCO_2RR generating various products. So the CO_2 transport process includes the quick-long $CO_{2,G,}$ and slow-short $CO_{2,D}$ diffusion and their conversion, that $CO_{2,D}$ diffusion is highly related to the reaction interface microenvironment of eCO_2RR.

In detail, longitudinal spatial distribution of $CO_{2,G}$ in the area of catalytic three-phase-interface (TPI) and cathode side of AZCBs with different interelectrode distances, is studied by COMSOL simulation in a potential range of 0.4−0.8 V vs. Zn/Zn^{2+} (Figure 3a; Figure S9). As expected, the AZCB exhibits the lowest $CO_{2,G}$ concentration at the cathode TPI, and that of pseudo-zero-gap battery (0.0775 cm) is lower than flow-type one (2.023 cm), suggesting enhanced $CO_{2,G} \rightarrow CO_{2,D}$ conversion rate. In addition, the lower concentration of $CO_{2,D}$ in the TPI area and anode side, suggests a faster $CO_{2,D}$ consumption rate in the pseudo-zero-gap AZCB, as compared with those with longer interelectrode distances (Figure 3b; Figure S10 and S11). Besides, the shorten interelectrode gap in batteries could also accelerate the electrolyte transmission at the CCL surface, therefore replenishing $CO_{2,D}$ at reactive CCL surface in time. According to Fick's law of diffusion, a steeper concentration gradient results in a higher diffusion flux. Therefore, the $CO_{2,G}$ and $CO_{2,D}$ diffusion efficiency in the zero-gap battery is higher than those in batteries with larger electrode gaps, due to the larger concentration gradient resulted from faster $CO_{2,G}$ conversion to $CO_{2,D}$ and $CO_{2,D}$ consumption by battery working.

Except CO_2 transport, ions shutting between the cathode and anode is also crucial. When the battery is discharging, concentration polarization will appear on the surface of the cathode and anode, that the OH^- concentration at the anode surface is higher than cathode due to electromigration. A reduction of the interelectrode gap leads to a concurrent increase in both the concentration (Figure 3c) and transport rate of OH^- ions (Figure S12) at the anode, demonstrating that the minimized gap design substantially improves the ion transport. Meantime, the distribution of OH^- concentration between 0.01 cm gap (zero distance) and 0.0775 cm (pseudo-zero-gap) are essentially identical, further demonstrating the rationality of the pseudo-zero-gap design in the experiment. According to Fick's law of diffusion ($J = -D^*dc/dx$), the higher concentration gradient indicates that the OH^- experiences minimal transport resistance between electrodes, leading to maximized mass transport efficiency, thereby increasing the effective concentration term in the Butler-Volmer equation and enhancing battery performance. Meanwhile, the anodic zinc dissolution rate (formation rate of Zn^{2+}) directly reflects the overall reaction rate of the battery. Similarly, the zero-gap battery exhibits in the highest zinc dissolution rate (Figures S13 and S14), which consequently leads to a rapid and vigorous reaction with OH^- [25, 26]. In total, the small electrode gap in AZCBs facilitates the ion transport, improving the battery performance and CO_2 electroconversion rate, therefore, creating a larger CO_2 concentration gradient, which benefits the CO_2 diffusion, and ultimately improving battery performance further.

In addition to mass transport, the eCO_2RR catalyst also affects the battery performance by regulating the discharging product, local reaction rate, etc. So eCO_2RR electrocatalysts are optimized by measuring their half-battery performance in a typical three-chamber flow-type electrolyser using GDEs in various media (6 M KOH and 0.5 M $KHCO_3$). As shown in Figure 3d, linear scan voltammetry curves (LSVs) in 6 M KOH clearly show that Bi_2O_3/C@350 displays higher current densities than the Bi_2O_3/C@400, Bi-hexahydroxytriphenyl (HHTP) MOF, and Bi_2O_3/C@250 catalyst. Chronoamperometry testing is also conducted, and thus steady-state polarization curves are obtained (Figure 3e; Figures S15 and S16). Accordingly, the eCO_2RR product distribution of as-prepared catalysts is also compared, where the formate Faradaic efficiency (FE_{HCOO-}) of Bi_2O_3/C@350 exhibits the highest value (up to 96.45%) in a wide current density range (Figure 3f–i), as compared to the Bi-HHTP, Bi_2O_3/C@250, and Bi_2O_3/C@400 samples. To ensure data reliability and reproducibility, we have conducted at least three independent parallel testing for FE_{HCOO-} of as-prepared catalysts, which is provided with error bars in Figure S17. Besides, eCO_2RR promoted by Bi_2O_3/C@350 also displays good performance even in a neutral electrolyte (0.5 M $KHCO_3$, Figure S18).

The charge transfer ability is also testified using electrochemical impedance spectroscopy (EIS) at −0.5 V vs. RHE, and corresponding equivalent circuits are simulated accordingly (Figure S19). Specifically, the eCO_2RR using Bi_2O_3/C@350 exhibits a lower charge transfer resistance (Rct, 0.85 ohm) than that of Bi-HHTP (1.83 ohm), while close system resistances (Rs, 1.0 vs. 0.91 ohm). The eCO_2RR process promoted by Bi_2O_3/C@350 is

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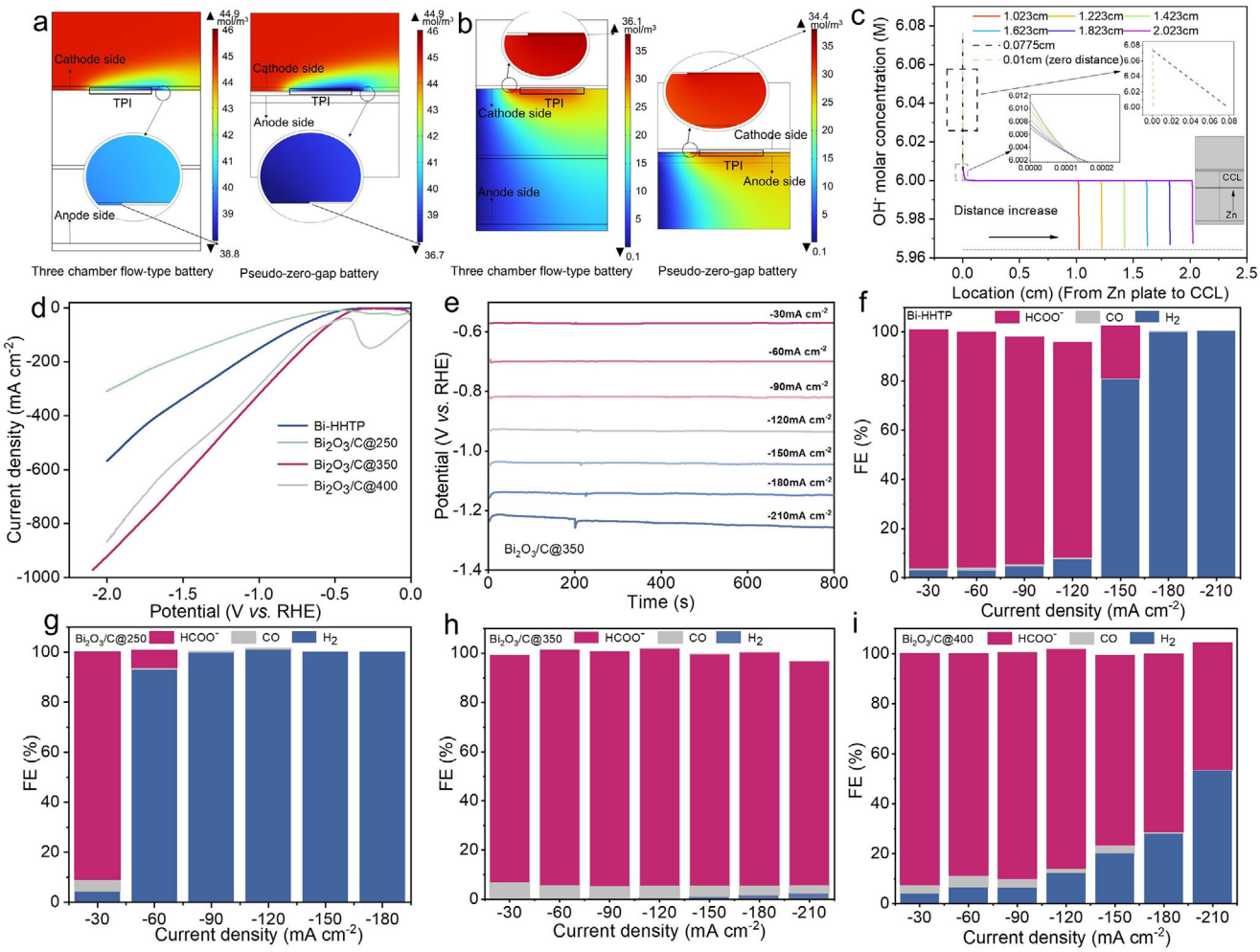

FIGURE 3 | (a) The gaseous CO_2 molar concentration fraction in the gas chamber and (b) dissolved CO_2 molar concentration at the TPI and anode side of AZCBs with distances of 2.023 cm (left) and 0 (right) at 0.4 V vs. Zn/Zn^{2+}; (c) The OH^- ion distribution along the center line of AZCB with different interelectrode distances (d) LSVs of half-battery eCO_2RR using different electrocatalysts (e) Chronoamperometry testing at different current densities of eCO_2RR using the Bi_2O_3/C@350 catalyst.; (f-i) FE values of eCO_2RR products promoted by Bi-HHTP, Bi_2O_3/C@250, Bi_2O_3/C@350, Bi_2O_3/C@400 catalysts.

also evaluated in a H-type electrolyser, whose Rs is much higher than that of the flow-cell, harming its eCO_2RR performance (Figure S20). Therefore, the above results confirm that the Bi_2O_3/C@350 catalyst displays efficient and selective eCO_2RR to formate performance, which is highly feasible in AZCBs.

2.4 | Catalyst Synthesis and Characterizations

Furthermore, the Bi_2O_3/C electrocatalyst is examined in detail, which is synthesized by annealing a bismuth-based metal organic framework (Bi-MOF) precursor made from $Bi(CH_3COO)_3$ salt and 3,6,7,10,11-HHTP (denoted as Bi-HHTP) ligand. By calcination of the Bi-HHTP MOF precursor at various temperatures (250 °C, 350 °C, 400 °C), as-obtained products are characterized using powder X-ray diffraction (PXRD, Figure 4a). First, XRD pattern of Bi-HHTP matches well with the reported literature, thus confirming its successful synthesis [27]. Main diffraction peaks of the sample achieved at 350 °C correspond to those of Bi_2O_3 (standard card PDF#78-1793), and a broad peak at 13° from carbon obtained by MOF annealing; the 400 sample also shows the same Bi_2O_3 crystal but bears extra peaks, suggesting over-annealing could result in impurities; while XRD of the 250 sample exhibits two distinct broad peaks possibly from carbon and Bi_2O_3 with low-crystallinity, indicating an insufficient calcination process. Therefore, the as-synthesized materials are labelled as Bi_2O_3/C@250, 350, 400, respectively, among which the 350 sample has well-defined crystallinity.

Next, the catalyst morphology is also characterized by using scanning (SEM), transmission electron microscope (TEM), and energy-dispersive X-ray spectroscopy (EDS) mapping images. As a result, Bi-HHTP shows a rod-like shape (Figure S21), and Bi_2O_3/C@350 displays a nanoparticle assembled rod shape, possibly due to Bi-MOF decomposition by calcination (Figure S22). In detail, the TEM and its EDS elemental mapping suggest homogeneous distribution of Bi, O, and C throughout Bi_2O_3/C@350 (Figure 4b), confirming the presence of carbon coating. Furthermore, its high-resolution TEM (HRTEM) images at various locations (Figure 4c; Figure S23) show lattice fringes with a d-spacing of 0.319 nm corresponding to the (201) facet of Bi_2O_3 crystals, and the amorphous carbon edges are present around

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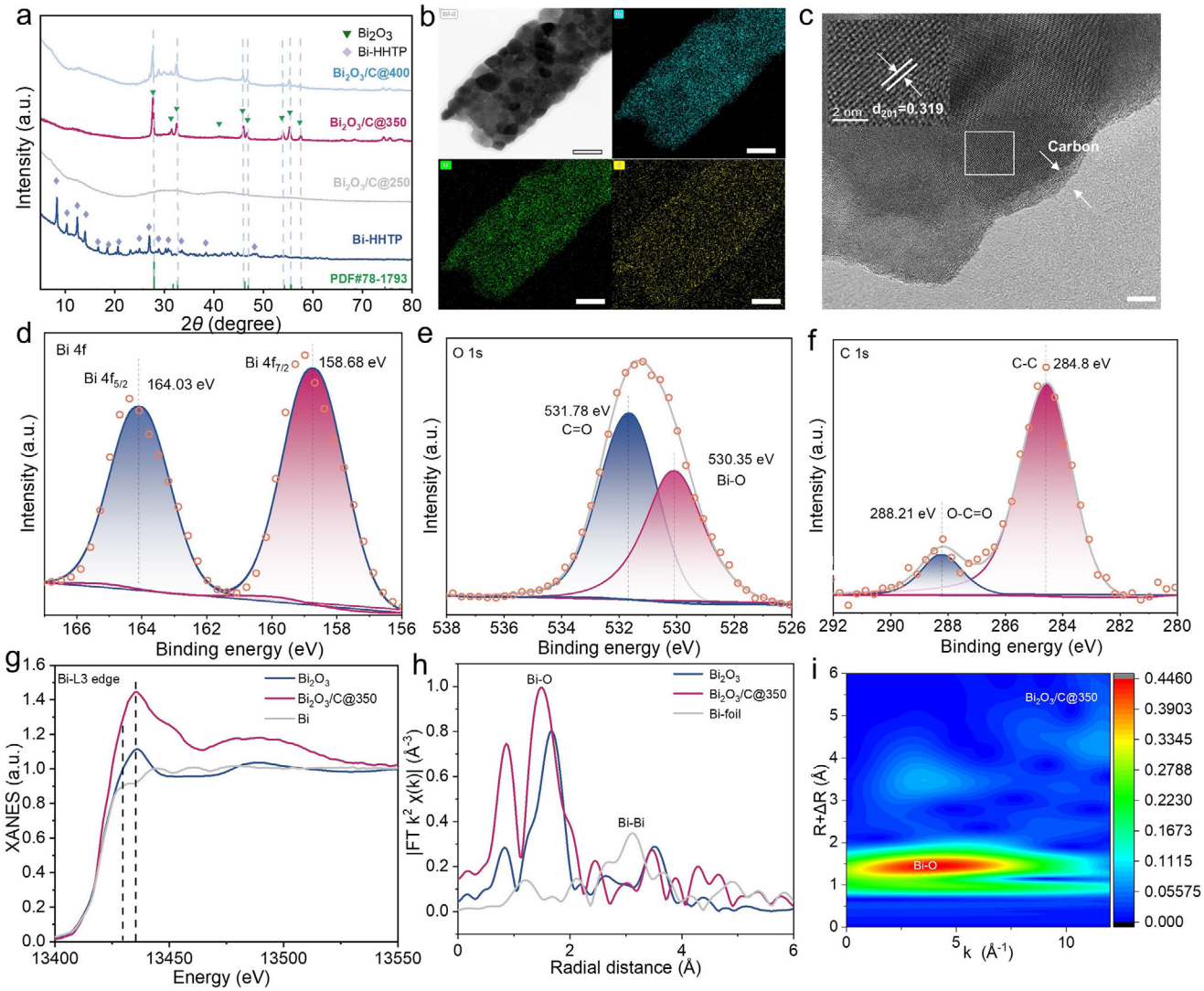

FIGURE 4 | (a) XRD patterns of electrocatalysts; (b) TEM and its EDS mapping of Bi, O, and C for Bi_2O_3/C@350 (scale bar: 50 nm). (c) HRTEM images of Bi_2O_3/C@350 (scale bar: 5 nm); (d–f) High resolution Bi 4f, O 1s and C 1s spectra of Bi_2O_3/C@350; (g) XANES of the Bi L_3-edge for Bi_2O_3/C@350, Bi_2O_3 and Bi foil reference.; (h) R-space of the Bi L_3-edge for Bi_2O_3/C@350, Bi_2O_3 and Bi foil reference; (i) Wavelet transform of the Bi L_3-edge of Bi_2O_3/C @350.

Bi_2O_3 nanoparticles. Selected area electron diffraction (SAED) also suggests that the synthesized samples are polycrystalline, in good agreement with the HRTEM result (Figure S24).

In addition, the element valence of electrocatalysts is measured using X-ray photoelectron spectroscopy (XPS, Figure 4d–f; Figures S25 and S26). The survey scan of Bi-HHTP MOF indicates the presence of Bi, O, and C, together with high-resolution Bi 4f, O 1s, and C 1s spectra confirming success synthesis of Bi-HHTP (Figure S25). Importantly, the survey scan of Bi_2O_3/C@350 also clearly shows the existence of Bi, O, and C elements (Figure S26). Especially, the high-resolution Bi 4f spectrum exhibits doublet peaks corresponding to Bi $4f_{5/2}$ (164.03 eV) and Bi $3f_{7/2}$ (158.68 eV), assigned to Bi^{3+} from Bi_2O_3 (Figure 4d) [28]. The O 1s doublet peaks located at 531.78 and 530.35 eV correspond to the C═O and Bi—O bond, respectively (Figure 4e) [29]. The main peak of C 1s spectrum (Figure 4f) at 284.8 eV is assigned to the C—C bond [30], and the satellite peak at 288.21 eV to the O—C═O bond, possibly because the carbon coating is oxidized by ambient air. Furthermore, the Bi_2O_3/C@350 after battery discharging has also been tested. First, post-XPS analysis shows a slight binding energy shift of Bi 4f peaks, indicating changes in the local electronic environment of Bi species after discharge, while the $-CF_2/-CF_3$ signals in C 1s and O—F components in O 1s originate from the PTFE-containing carbon paper substrate (Figure S27). Then post-XRD pattern of the electrode largely retains the characteristic diffraction peaks of Bi_2O_3/C@350, suggesting that the crystal structure remains stable during the discharge process (Figure S28).

The synchrotron X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) of Bi_2O_3/C@350 have also been analyzed (Figure 4g–i; Figures S29 and S30). XANES spectrum of Bi_2O_3/C@350 exhibits an adsorption edge in the range of 13420–13430 eV, which is close to that of Bi_2O_3 reference, and higher than that of Bi foil (Figure 4g), indicating Bismuth is present as Bi^{3+}. Additionally, the R-space of Bi_2O_3/C@350 exhibits a peak around 1.50 Å

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attributed to Bi-O coordination, close to that of Bi_2O_3; while that of Bi foil shows a peak around 3.09 Å assigned to Bi—Bi bonds. And the wavelet transform (WT) contour map of EXAFS at the Bi L_3 edge (Figure 4i) reveals that Bi_2O_3/C@350 exhibits a maximum intensity at approximately 1.50 Å, consistent with the R-space data [31]. Besides, Raman spectroscopy further confirms the presence of carbon shell and Bi_2O_3 nanoparticles (Figure S31). The thermogravimetry (TG) and its differential scanning calorimetry (DSC) curves of Bi-HHTP further suggest that proper annealing could result in a carbon-coated bismuth oxide (Figure S32).

2.5 | DFT Simulations of eCO_2RR Mechanism

DFT calculations have also been conducted to inspect the mechanism of half-battery-reaction eCO_2RR catalysed by as-prepared materials. The specific structure models of Bi-HHTP and Bi_2O_3/C@350 are displayed in Figure S33. According to experimental results, for the eCO_2RR to monocarbon product process, the CO and HCOOH pathway are highly competitive with each other. The key is the adsorption structure of CO_2 on reactive Bi sites, that C@Bi adsorption usually leads to $*CO_2 \rightarrow *COOH \rightarrow *CO$, while O@Bi adsorption results in $*OCO \rightarrow *OCHO \rightarrow *OCHOH$ [32]. Under pH = 0 media, the first proton-coupled electron transfer (PCET) step for both the CO and HCOOH pathway is endothermic for Bi-HHTP and Bi_2O_3/C@350 catalyst (Figure 5a; Figure S34a). While the second PCET step is exothermic for both reaction pathways of these two samples. Therefore, the first PCET step is the rate-determining step (RDS), that is $*CO_2 \rightarrow *COOH$ and $*CO_2 \rightarrow *OCHO$ for the CO pathway and HCOOH pathway, respectively.

To enhance the credibility of DFT simulations, pH correction is applied to the free energy diagram, in accordance with the experimental condition (pH 14.78, Figure 5b; Figure S34b) [33]. For the Bi_2O_3/C@350 sample, the Gibbs free energy change (ΔG) of RDS (ΔG_{RDS}) for the CO pathway (1.08 eV) is much higher than that for the HCOOH pathway (−0.69 eV), suggesting its preference to generate HCOOH. The Bi-HTTP also prefers to generate HCOOH product, because of the lower ΔG_{RDS} (0.93 eV of the HCOOH pathway vs. 1.38 eV of the CO pathway). Besides, for the HCOOH pathway, ΔG_{RDS} of Bi_2O_3/C@350 (−0.69 eV) is much lower than that of Bi-HTTP (0.93 eV), suggesting its advantageous eCO_2RR to HCOOH performance. Accordingly, selectivity Pourbaix diagrams of Bi_2O_3/C@350 and Bi-HHTP are calculated (Figure 5c,d), that a much larger gap in ΔG_{RDS} between the CO and HCOOH pathway for Bi_2O_3/C@350 than that of Bi-HTTP, further confirms its great HCOOH selectivity [34]. Specifically, the adsorption energy of eCO_2RR intermediates on Bi_2O_3/C@350 and Bi-HTTP catalyst is compared, suggesting Bi_2O_3/C@350 shows much higher formate performance over CO, and over that of Bi-HTTP (Figure 5e,f).

To understand underlying reasons for eCO_2RR performance, charge density difference and Bader charge analysis are conducted (Figure 5g; Figure S35 and Table S2), which demonstrates unambiguously charge transfer between reactive Bi atoms and adsorbates. When *OCHO and *COOH intermediates are present on Bi_2O_3/C@350 and Bi-HHTP, electron accumulation regions are observed, suggesting facile adsorption between intermediates and active sites benefiting eCO_2RR [35]. Bader charge analysis indicates that Bi atoms lose electrons while coordinating O atoms gain electrons [36]. In case of the HCOOH pathway, the *OCHO adsorption results in great electron loss by Bi atoms, and the adsorbed O atom in *OCHO gains the greatest number of electrons for both Bi_2O_3@C (-1.19 e) and Bi-HHTP (-1.06 e), meaning stronger interactions. In contrast, the CO pathway has also been analysed, that the number of electrons transferred between *COOH and Bi_2O_3/C@350 (1.14 e) is less than Bi-HHTP (1.30 e), indicating *COOH adsorption and thus the CO product is not favorable. Finally, projected density of states (PDOS) of Bi and O atoms in Bi_2O_3/C@350 exhibit a significant overlap below the Fermi level (Figure 5h,i). This also proves that charge transfer occurs between Bi and O, therefore, there are strong interactions between Bi and *OCHO. The PDOS overlap of Bi and C atoms is slightly less than that between Bi and O atoms, suggesting that the interaction between Bi-*COOH is weaker than that between Bi-*OCHO. This is consistent with experimental results, that both materials exhibit attenuated interactions with *OCHO and thus higher selectivity for the HCOOH than CO product.

As shown in Figure S36a,b, the in situ FTIR spectra exhibit the emergence of characteristic absorption bands at 1400 and ~1350 cm^{-1} under increasingly negative potentials, which can be assigned to the *OCHO intermediate and CO_{3-} species, respectively [37]. The formation of *OCHO intermediates is a well-recognized signature of the formate production pathway. Notably, no detectable *CO (~2100 cm^{-1}) [38] adsorption bands were observed, suggesting that the CO evolution pathway is suppressed on Bi_2O_3/C@350. On the other hand, in situ Raman spectra is also collected at various potentials (Figure S36c), showing distinct peaks at ~1450 cm^{-1} corresponding to *OCHO intermediates, and at ~1650 cm^{-1} assigned to HCO_3^- species [39]. The increasing intensity of the*OCHO-related band at potentials below −0.8 V further confirms that *OCHO is the dominant adsorbed intermediate on the Bi_2O_3/C@350 surface. These combined operando FTIR spectroscopic observations provide direct experimental evidence consistent with our DFT calculations, which indicate that Bi_2O_3/C@350 exhibits stronger adsorption and better stabilization of the *OCHO intermediate than *CO (~1900 cm^{-1}) [40].

2.6 | Broad Implications of Pseudo-Zero-Gap AZCBs

Except the above-mentioned Bi_2O_3/C@350 material, other as-prepared catalysts have also been blended into this zero-gap battery (Figure S37), that Bi-HTTP, Bi_2O_3/C@250, Bi_2O_3/C@400 shows a peak power density of 26.1, 25.6, and 54.4 mW cm^{-2}, respectively, which are also higher than reported data (Table S1). Significantly, the product distribution of pseudo-zero-gap AZCB using Bi_2O_3/C@350 during discharging is measured (Figure 6a), that FE_{HCOO-} is above 80.0% over all discharging current densities, with the highest value of 87.0% at 4 mA cm^{-2}. This CO_2 electroconversion performance is better than that of the flow-type AZCB, which shows an FE_{HCOO-} ranging 75%−78% during battery discharging in the current density range of 1–6 mA cm^{-2} (Figure S38). The catalyst loading amount onto the GDL is also optimized as 5 mg cm^{-2} (Figure S39). Except the Bi_2O_3/C cathode, we measure the battery performance of commercial Ag powder and

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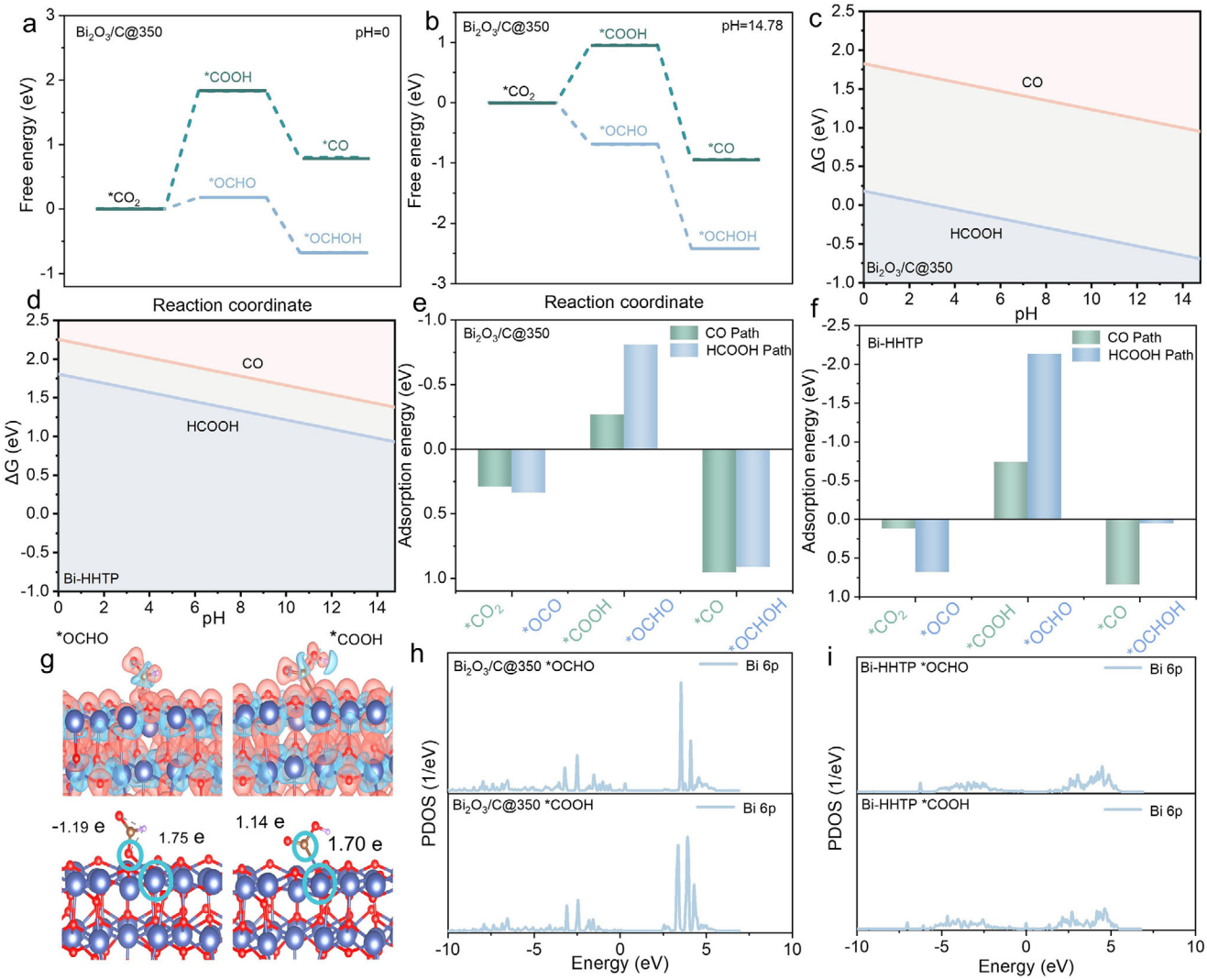

FIGURE 5 | (a,b) Free energy diagrams for the eCO_2RR to CO and HCOOH pathway in media with pH 0 and 14.78 using Bi_2O_3/C@350 catalyst.; (c-d) The eCO_2RR selectivity Pourbaix diagram for Bi_2O_3/C@350 and Bi-HHTP; (e,f) Adsorption energy of the *COOH and *OCHO intermediate for Bi_2O_3/C@350 and Bi-HTTP; (g) Difference charge density and Bader charge analysis of the *OCHO and *COOH intermediate for Bi_2O_3/C@350; (h,i) Projected density of states (PDOS) of Bi 6p orbitals for Bi_2O_3/C@350 and Bi-HTTP.

Bi-TDC (2,5-thiophenedicarboxylates) metal–organic framework (MOF, which also generates formate and is developed previously by our group [34]) to further demonstrate the advantageous performance and broad applicability of such pseudo-zero-gap AZCBs. As shown in Figure S40, the pseudo-zero-gap batteries exhibit consistently higher discharge-potential plateaus across all tested catalysts, as compared to the H-type and flow-type batteries. Moreover, the discharge current densities of pseudo-zero-gap AZCBs are several times greater than that of the H-type and flow-type one (Figure S41). These results further confirm the facile CO_2 electroconversion ability of such AZCBs, together with excellent electricity-generating capability, outperforming traditional battery architecture.

To demonstrate the practical application potential of such pseudo-zero-gap AZCBs, stability is improved by using a layer consisted with ZrO_2, carbon powder, and polyvinylidene fluoride in a mass ratio of 8:1:1 onto the Zn anode (Figure S42) [41]. As shown in long-term galvanostatic charge-discharge cycling (Figure 6b), the AZCB using a modified anode operates stably over 600 cycles under a constant current density of 0.5 mA cm^{-2}, much longer than that uses bare Zn, possibly because ZrO_2 nanoparticles provide uniform nucleation sites promoting homogeneous Zn deposition. Besides, the ZrO_2-modified Zn anode shows stable charge/discharge potential platforms (Figure S43). As shown in Figure 6c, the battery performance of the ZrO_2-Zn anode is also evaluated with a peak power density of 63.7 mW cm^{-2}, comparable to that of the bare-Zn-anode battery (67.0 mW cm^{-2}), indicating no detrimental impact on battery performance, although improved durability. Key performance metrics are also compared between the pseudo-zero-gap and the flow-type battery (Figure 6d), confirming the advantageous performance of the pseudo-zero-gap AZCB. Besides, it exhibits an impressive specific capacity of 4534 mAh g^{-1} at a high current density of 500 mA g^{-1}, which still maintains 3036 mAh g^{-1} at 1000 mA g^{-1} (Figure S44), assigned to excellent ion transport and charge/discharge capability of the fabricated battery. Importantly, the peak power density of pseudo-zero-gap AZCB is the highest

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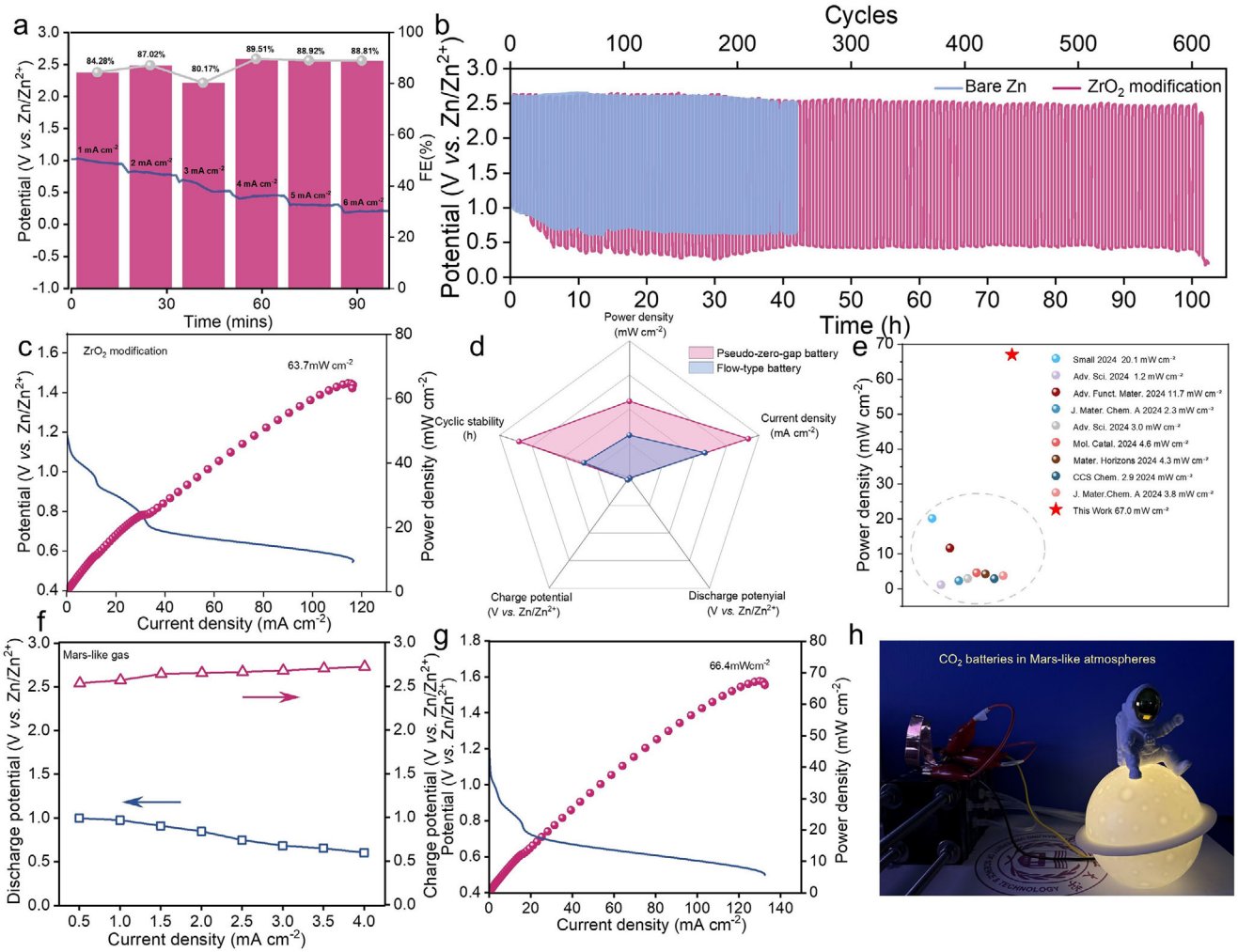

FIGURE 6 | (a) Constant discharge curves and Faraday efficiency of HCOOH for pseudo-zero-gap AZCB with Bi_2O_3/C@350; (b) Charge and discharge cycling over long-term operation; (c) The *I–V* and *I–P* curves of the ZrO_2-Zn battery, (d) The battery performance comparison of the pseudo-zero-gap and flow-type AZCB. (e) The power density comparison of this work and reported data. (f) Continuous charge and discharge curves of the above AZCB in the Martian atmosphere. (g) The *I–V* and *I–P* curves of the above AZCB in the Martian atmosphere. (h) A battery stack using three AZCBs working in the Martian atmosphere. The electrolyte is 6 M KOH.

as compared to the reported value (mostly less than 10 mW cm^{-2}, Figure 6e), demonstrating that AZCB with advantageous configuration and facile electrocatalyst achieves a ground breakthrough in battery performance.

Significantly, the performance of pseudo-zero-gap AZCB is also measured in a simulated Martian atmosphere (95.0% CO_2 and 5.0% N_2), to explore its potential for Mars exploration. As depicted in Figure S45, the pseudo-zero-gap battery exhibits a relatively stable OCP around 1.38 V in the Martian atmosphere, close to 1.48 V of the pure CO_2. Continuous charge-discharge performance curves are also tested, which show comparable battery performance with that under pure CO_2 (Figure 6f). In detail, the peak power density is 66.4 mW cm^{-2}, also close to that of a pure CO_2 atmosphere (67.0 mW cm^{-2}). To further validate the practical application, three pseudo-zero-gap AZCBs are assembled into a series-connected stack to light a prototype space explorer (4.5 V, Figure 6h) and bulbs (Figure S46) under the Martian atmosphere, exhibiting long-lasting and stable brightness. However, the atmospheric pressure of Mars is only about 0.6% of Earth, significantly lowering the partial pressure of CO_2 (from 0.95 to 0.0057 atm) despite its high molar fraction (~95%). Therefore, the battery performance of pseudo-zero-gap AZCB is also tested in a 1%-CO_2 atmosphere, displaying a peak power density of 0.17 mW cm^{-2} (Figure S47a). Even though this performance is so good, Earth-like working senecios are also applicable (Figure S47b), for example, extracting the Martian gas to a Human-Mars explorer. This renders AZCB to work under the ambient pressure (1 atm), while still offers practical solutions for generating carbon-based feedstocks and electric energy in future Mars exploration.

3 | Conclusion

In summary, a AZCB with a pseudo-zero interelectrode distance is designed and fabricated using a glass-frit separator between the Zn anode and cathode constructed by Bi_2O_3/C@350 GDE. This near-zero-gap AZCB displays much better battery performance than the H-type and flow-type battery, with the largest peak

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power density of 67.0 mW cm^{-2}, a discharge potential of 0.57 V, and a current density of 117.5 mA cm^{-2}, together with robust cycling performance over 600 cycles. Such an advantageous battery performance of as-fabricated pseudo-zero-gap AZCB originates from promoted mass transport of hydroxide ions and CO_2, and the efficient and selective CO_2 electroconversion ability of Bi_2O_3/C@350 catalysts. Significantly, the minimized electrode gap could improve the hydroxide ion transport and thus improve the battery performance and CO_2 consumption rate at the reactive interface, therefore, leading to a large concentration gradient that benefits the $CO_{2,D,}$ and $CO_{2,G}$ diffusion and their conversion. This, in turn, greatly improves the battery performance further, which is a bilateral mechanism between battery performance and mass transport. Interestingly, such zero-gap AZCBs also exhibit great performance under Mars-like atmosphere, providing a highly promising energy and feedstock solution for Mars exploration. Therefore, this work proposes an innovative battery configuration for AZCBs, breaking the performance limit of traditional H-type structure, greatly promoting the lab-scale research to practical applications for aqueous CO_2 batteries not only on Earth but also on Mars.

Author Contributions

Weiliang Li was responsible for methodology, data curation, and writing the original draft. Guoliang Xu and Qi Huang contributed to data curation and validation. Tengda Lu performed data analysis. Qiang Li and Sheng Chen contributed to conceptualization and writing – review and editing. Jingjing Duan was responsible for conceptualization, supervision, and writing – review, and editing.

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Conflicts of Interest

Jingjing Duan has filed Chinese provisional patent applications (No. 2025104745577) based on this work. Other authors declare no competing interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

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